

Serein Maison

Secure E-Commerce Engine

Architecture, security, payment flow, and verification report · 2026

A premium boutique storefront for considered leather goods, built with React, Express, MongoDB, Mongoose, JWT authentication, bcrypt password hashing, and Stripe test mode.

Architecture

The application is split into independent client and server npm workspaces. React and Vite provide the storefront, while Express and Mongoose provide the API and persistence boundary. Models cover users, products, carts, and orders. Routes separate authentication, catalog, carts, checkout, and order history.

•Te

n-p

rod

uct

id

em

pot

ent

M

on

go

DB

ca

tal

og

se

ed

•J

W

T s

es

sio

ns

wit

h p

rot

ect

ed
car
t, c
he
ck
out
, a
nd
ord
er
rou
tes
•St
rip
e
Ele
me
nts
pa
ym
ent
for
m
wit
h s
erv
er-
sid
e
Pa
ym
ent
Int
ent
pri
cin
g
•R
es
po
nsi
ve
edi

tori
al
UI
wit
h s
kel
eto
ns,
e
mp
ty
sta
tes
, to
ast
s,
an
d r
ed
uc
ed-
mo
tio
n s
up
por
t

Security Decisions

•P
as
sw
ord
s u
se
bcr
ypt
js
wit
h
a
wo

rk f
act
or
of
12.
•J
W
Ts
us
e
a s
ev
en-
da
y e
xpi
ry,
iss
uer
, a
nd
cur
ren
t-u
ser
lo
ok
up.
•R
egi
str
ati
on
val
ida
tes
e
ma
il a
nd
str
on
g p

as
sw
ord
s s
erv
er-
sid
e.
•H
el
me
t,
C
O
RS
all
ow
listi
ng,
bo
dy-
siz
e li
mit
s,
AP
l ra
te l
imi
tin
g,
an
d a
uth
rat
e li
mit
ing
ar
e e
na
ble
d.

•St
rip
e s
ecr
et
ke
ys
re
ma
in
ser
ver
-on
ly;
the
br
ow
ser
re
cei
ve
s o
nly
a
pu
bli
sh
abl
e-k
ey
co
nfi
g v
alu
e.

•P
ay
me
ntl
nte
nt
ow
ner

shi
p
is
ma
tch
ed
to t
he
aut
he
nti
cat
ed
us
er.
•Pr
ice
s,
sto
ck,
ca
rt t
ota
ls,
an
d p
ay
me
nt
am
ou
nts
ar
e a
lw
ay
s r
ec
alc
ula
ted

on

the
se
rve
r.

Payment Flow

•1.

Aut
he
nti
cat
ed
ch
ec
ko
ut l
oa
ds
the
cu
rre
nt
car
t fr
om
M
on
go
DB
.

•2.

Th
e
AP
I cr
eat
es
a
Str
ipe
Pa
ym

ent
Int
ent

for
the
da
tab
as
e s
ubt
ota
l.

•3.
Str
ipe
El
em
ent
s c
onf
irm
s t
he
tes
t p
ay
me
nt
in t
he
bro
ws
er.

•4.
Th
e
AP
I re
trie
ve
s t
he

Payment
entire
and
accepted
only
success
deduct
dis
tats.
•5.
Stock
is
decre
ment
ed, the
order
is r
ecord
ed,
and
the
car
t
is
clear
are

d
in
a
Mo
ng
o tr
an
sa
cti
on.

Stripe Test Cards

•S

uc

ce

ss:

42

42

42

42

42

42

42

42

•G

en

eri

c d

ecl

ine

: 4

00

0 0

00

0 0

00

0 0

00

2

•

3D

Se

cur
e:
40
00
00
25
00
00
31
55

Use any future expiry date, any three-digit CVC, and any valid postal code.

Verification

All 8 automated tests pass across API and client, covering API health and safe errors, bcrypt registration and password validation, cart total representation, database-derived payment amounts, foreign PaymentIntent rejection, and React rendering. The live journey was completed against Atlas and Stripe test mode: a new account registered, two products (three units) were added, Stripe confirmed USD 1,590.00 as succeeded, Atlas recorded a paid order, cleared the cart, and reduced stock by the purchased quantities. Sign-out/sign-in preserved order history. The 375px, 768px, and 1440px sweep covered the active Payment Element and all store routes without horizontal overflow. Checkout waits for authenticated cart hydration and initializes Stripe Elements once under React StrictMode.

Design System

Ivory paper, deep forest green, oxidized brass, and restrained rust accents create the boutique direction. Cormorant Garamond provides editorial personality; DM Sans keeps operational UI legible. Product imagery and generous spacing are paired with practical mobile navigation and responsive layouts at 375px, 768px, and 1440px.